

Statistical Language Acquisition and its Implications for Universal Grammar

Introduction and Literature Review

1. **Research Topic:** A core concept in the genesis of cognitive science is that of the innateness of language, and the primary area of inquiry regarding this innateness is that of language acquisition. All children, regardless of which spoken or signed language they are exposed to early in life, generally develop the language faculty uniformly, advancing from babbling in infancy to single word, multiple word, and more sophisticated stages of language use.
2. **Importance in Science:** The issue of language acquisition is important in science because the method of acquisition holds ramifications for human learning in general – these methods being purely statistical acquisition or a combination of innate and statistical acquisition. If language acquisition has an indispensable innate component, then that means that other mental faculties could as well. If that is the case, then scientific concepts like artificial general intelligence could be proved impossible (or, at least, impossible using statistical methods as used in current machine learning algorithms).
3. **Application of Course Content:** B.F. Skinner's theory of operant conditioning holds that children learn language statistically from their environment, that language acquisition is nothing more than a set of associations between words and things which are then imitated by the child and reinforced by their parents or other caregivers. According to this theory, a computer could easily replicate human language use with a corpus of similar linguistic data. However, Noam Chomsky's nativist theory attempted to refute Skinner's ideas with the argument that language cannot simply be a set of responses to external stimuli, since language users can create entirely novel sentences that they have never heard before, creating an infinite set of linguistic output out of a finite set of words. Chomsky holds that humans have an innate language acquisition device, which supports our use of universal grammar and allows us our normal, creative acquisition and use of language. He argues that we must have this acquisition device because our environments don't provide enough linguistic data to teach us the syntactic rules governing normal language use. Steven Pinker, finally, incorporates Chomsky's ideas of innateness (and its importance for understanding the human brain), but emphasizes that particular languages are learned from one's environment using a combination of universal grammar and experience.
4. **Problem Statement:** It is currently unclear if language acquisition could be purely experience-dependent – that is, if the rules of language could be learned statistically from one's environment or if these rules require an innate, universal grammar. While use of a language's specific vocabulary is experience-dependent, it's not clear if the syntactic rules governing all language use could be learned from one's environment or are solely innate within humans.
5. **Hypothesis:** I hypothesize that human acquisition of the core structure of language is innate and is something that a computer is not able to do using statistical methods. If a machine learning algorithm were tested on a training set similar in robustness to that of a child's exposure to language and attempted to recreate a child's creative use of language, I believe that it would fail to match the linguistic sophistication of a ten-year-old child.

6. Operationalize Terms: Language acquisition device: the biological human capacity to acquire and use language. Syntax: the structure of coherent sentence formation in language. Universal grammar: the postulate that humans have a set of structural rules of language formation innate to them, and that all languages share this universal grammar. Poverty of the stimulus: the idea that a child's environment does not provide enough data to teach a child the syntactic rules governing language use. Statistical language acquisition: the acquisition of language's vocabulary and grammatical structure from one's environment.
7. Literature Review: Judy Kegl and Gary Morgan (Kegl) performed a study on some members of a famous school for deaf children in Nicaragua. In this school, children who had not been exposed to sign languages invented their own language, Nicaraguan Sign Language (ISN), from very little external input. Children who had first invented the language created a language with simpler grammatical rules than later children, who learned ISN and expanded it with more sophisticated syntax into a more complex language. Kegl and Morgan performed a study on 22 of these children and now-adults, who had acquired ISN at different times in its development as well as at different ages, in which the participants were given a false-belief task (which measured their ability to attribute mental states to cartoon characters). The study showed that language acquisition by the age of 10 was correlated with higher scores on the false-belief task. This study, then, implies that language acquisition, whether innate or externally acquired, correlates with the ability to perform philosophical tasks such as attributing minds to other agents in the world. Jennie Pyers and colleagues (Pyers) performed a separate study on children from the same school, who had acquired ISN at different stages in its development, which showed that acquisition of the more complex form of ISN was correlated with higher degrees of spatial cognition when performing spatially guided searches. These results imply that acquisition of a sufficiently complex language is vital for spatial cognition, but they are limited in that they do not show whether the specific vocabulary of a language allows for spatial cognition or if it is the expression of an innate language faculty that allows for spatial cognition. The fact that spatial cognition is vital for survival supports the idea of universal grammar, since universal use of spatial language would have helped humans survive up to the present moment. Each of these studies, since they were performed on people who had not been exposed to language as young children, emphasize the importance of language acquisition in other cognitive tasks and provide some support for universal grammar, since the lack of a universal grammar in early humans would have been detrimental to the survival of the species to the present moment. However, the studies do not prove the existence of universal grammar or its role in language acquisition in comparison with statistical methods of language acquisition. Since it is so difficult to study universal grammar itself, one must look to studies of statistical language acquisition methods to clarify its role in language acquisition. Jenny Saffran and colleagues (Saffran) performed a study on infants in which they were exposed to two minutes of unsegmented fluid speech of a made-up language as a familiarization set and were later exposed to two test sets. One set was made up of three-syllable words with word boundaries derived from the familiarization set, and another was made up of very similar items but with different

word boundaries. The only clues that the infants had as to which words should have been more familiar were the probabilities of occurrence between pairs of syllables in the familiarization set. The results, measured using the infants' visual fixation level on a blinking light, showed that the infants had longer listening times for the words that were not familiar to them, implying that the infants were able to recognize the difference between new and familiar orderings of the strings of syllables after only two minutes of listening. These results demonstrate statistical acquisition of some aspects of language use, but this study was limited in that it was unclear if the infants were performing actual language acquisition or simply performing a general learning mechanism in which patterns were gleaned from their environment. The study, then, did not rule out universal grammar as a part of language acquisition, it merely showed that statistical learning could play a role in the acquisition of vocabulary. Charles Yang (Yang), finally, as part of his paper comparing statistical and universal grammar approaches to language acquisition, performed a study in which he trained a machine learning model on a familiarization set inspired by Saffran and colleagues', and then used that model to predict word boundaries the same way that the infants did in the prior experiment. The results found that the machine learning model did not perform as well as the infants in terms of its familiarity with the words, emphasizing the need for other human faculties of language acquisition in word recognition. While refuting the idea that a language could be learned purely by learning probabilities of syllables, Yang's study did not address whether a language's grammar could be learned statistically and therefore did not prove that language itself could not be learned statistically. A study of a machine learning model that can learn a language's grammar statistically and that is exposed to a similar corpus of linguistic data as a child is needed to further clarify whether it is possible for a human to learn a grammar statistically, and, hence, whether a human can acquire language statistically.

8. Ethical Implications: Language acquisition has always been difficult to study ethically because the only straightforward method of studying it would be to deprive a child of their normal linguistic environment. The development of Nicaraguan Sign Language provided an exception, and results from studies of its users provide support for universal grammar. Cases of neglected children who were not exposed to language are problematic in the study of language acquisition because the lack of an otherwise healthy environment could be at play in their lack of language use. Regardless, studies focused on statistical language acquisition have been more robust because of the lack of ethical concerns in studying it and have therefore shed more light on the innateness of language. Turning to the study posited here, the main method of data collection would be the collection of sound recordings from children's environments, making privacy a priority in the collection and storage of this information and encryption a key tool in ensuring this privacy.

Methods

9. Participants: The participants of this study would include human children from ages 0 up until age 10 (when prior studies have shown that language acquisition begins to wane),

who can hear and use spoken language (since the study will utilize transcriptions of audio recordings), as well as the adults who interact with them (including parents and teachers). Children who primarily use American Sign Language or other sign languages will unfortunately be excluded due to the materials used in the study. The children need to be in this age range because the acquisition of language wanes beginning around age 10, and it is this childhood acquisition that Noam Chomsky's theory of universal grammar refers to (not acquisition later in life or that of a second language).

10. **Materials:** Materials would include microphones to record the spoken language the children are exposed to, as well as computer servers to run algorithms to process the data.
11. **Technology:** The technologies used in the study would be a transcription algorithm to transcribe the audio recordings into text (such as that available from Google), and a machine learning model that is designed to statistically learn vocabulary and grammar, and that is trained on the output of the transcription algorithm. An online terminal would be used to interact with the machine learning model after training and test its use of language.
12. **Design:** Several schools, preschools and daycares would be contacted to take part in the study, as well as parents of infants and younger children. After the subjects' parents agreed to have their children participate in the study, recording would begin in the classrooms during certain hours of the day, and the parents of the infants and younger children would start being called in to approximate their at-home interactions with and around their children. All language spoken would be recorded and stored in encrypted files on secure servers and accessed only by research technicians involved in the study. Enough interactions would be recorded as to approximate the linguistic material a single child is exposed to from age 0 to 10 – for example, the same number of classroom settings a child experiences in real life would be recorded in the study, though not involving the same child or same classroom setting. Earlier experiences of language exposure are difficult to quantify, so the number of interactions recorded involving younger children will be arbitrary but ample.
13. **Procedures:** The classroom settings that are recorded would operate as if no study were occurring, with the participants not changing their behavior at all, and the younger children and parents will approximate their at-home experiences – doing things like reading to the child, talking about their days, and speaking “parentese” to the child (speaking simply and concisely and emphasizing content words). Once all the recordings are collected, they will be transcribed by a transcription program and used to train a specialized machine learning model that employs a statistical algorithm like k-nearest neighbors. The machine learning model would be designed to extract not only statistical information about things like word order and vocabulary but also syntactic rules governing the grammatical use of language. Finally, a user will interact with the output of the model via an online terminal and direct it to write paragraph-length responses to prompts which will then be compared to real responses from ten-year-olds and graded by a third party to determine if these responses approximate the linguistic sophistication of a ten-year-old child.

Data Plan

14. Data: The independent variable, which would be the input to the machine learning model (the transcription of audio recordings), would affect the dependent variable, which would be in the form of textual output from the machine learning model. This textual output would be the data that would support the hypothesis. Specifically, the output will be its responses to simple prompt questions. A confounding variable could be that of the transcription algorithm incorrectly transcribing words.
15. Data Analysis: The data in the form of prompt responses from the model would be graded in comparison with prompt responses from real ten-year-old children by a third-party grader according to a standard rubric (who does not know which are the machine's responses), and the response scores from a large sample of machine responses and controls would be averaged and compared. To show support for the hypothesis, the machine would score lower on average than the children, even though the independent variable (language exposure) would have been similar for the algorithm as for the child response writers.
16. Data Interpretation: Theoretically, if humans acquire language simply through statistical means as B.F. Skinner posited, then a computer algorithm would be able to approximate human language use given a similar corpus of linguistic information to imitate (especially if that algorithm extracted not only word choice statistically but also syntactic structure). If the data from this study confirmed the hypothesis (that is, if the algorithm scored lower on average on its prompt responses), the results would support Noam Chomsky's nativist theory of language acquisition, in which the grammatical structure of language could not be learned solely by statistical means and would require an innate language acquisition device that is currently unique to humans.

Conclusions

17. Conclusions: It's been well established that vocabularies of specific languages are acquired using statistical means, but the idea of a universal grammar to explain the creative use of language out of a poverty of stimulus has made sense theoretically but not been proven scientifically. The most straightforward way to prove universal grammar's existence would be to prove that the syntactic rules governing normal language use could not be learned from the limited data in a child's environment, and not by purely statistical means. This study would attempt to make some headway in that endeavor, by proving the hypothesis that human language use could not be recreated using current machine learning technology due to the statistical method of their operation (even when using a similar corpus of data as a human). These results would support nativist theory and cognitive science's emphasis on innate human faculty in general - just as recent literature like Yang's paper does - and it would cast doubt on the viability of concepts like artificial general intelligence and the idea that human cognition could be recreated in silicon, perhaps pushing technology in a direction that is more focused on supporting humanity as opposed to approximating it.

18. Limitations: The study would be working only with current machine learning technology, and it's not clear if another technology could arise in artificial intelligence that is less dependent on data and could better approximate innate human faculties. The data that would be fed into the model are also a rough approximation of the linguistic data that a child would experience.
19. Future Direction: The area in cognitive science of language acquisition could be further explored using computational methods that are not statistically based. There could, in fact, be a computational basis for universal grammar that could be approximated by a computer using algorithms that haven't been developed yet. The important point would remain, however, that language acquisition in humans is probably based on an innate, experience-independent faculty, whether this could be approximated by a computer or not.

References

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